

Linear models: recursive least squares (RLS)

Standard least squares: $f(\mathbf{x}_n) = \mathbf{x}_n^\top \mathbf{w}$, loss function $(f(\mathbf{x}_n) - y_n)^2$, covariance matrix $\Sigma_n = X^\top X$, $X \in \mathbb{R}^{n \times d}$. The solution for $f^*(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}^*$ is

$$\mathbf{w}^* = (X^\top X)^{-1} X^\top y = \Sigma_n^{-1} \sum_{i=1}^n \mathbf{x}_i y_i$$

RLS: based on the matrix inversion lemma:

Initialize $\mathbf{w}_0 = 0 \in \mathbb{R}^d$, and $\Gamma_0 = I \in \mathbb{R}^{d \times d}$.

Iteration:

$$\begin{aligned}\Gamma_n &= \Gamma_{n-1} - \frac{\Gamma_{n-1} \mathbf{x}_n \mathbf{x}_n^\top \Gamma_{n-1}}{1 + \mathbf{x}_n^\top \Gamma_{n-1} \mathbf{x}_n} \\ \mathbf{w}_n &= \mathbf{w}_{n-1} - \Gamma_n \mathbf{x}_n (\mathbf{x}_n^\top \mathbf{w}_{n-1} - y_n)\end{aligned}$$

It can be shown that $\Gamma_n = \Sigma_n^{-1}$.

The regularized version is obtained as usual by introducing a regularizer in the loss function, $\sum_{i=1}^n (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 + \lambda \|\mathbf{w}\|_2^2$, and starting from $\Gamma_0 = (I + \lambda)^{-1}$. This converges to $\Gamma_n = (\Sigma_n + \lambda I)^{-1}$.

Stochastic Gradient Descent

Instead of updating w_n using

$$w_n = w_{n-1} - \Gamma_n x_n (x_n^\top w_{n-1} - y_n)$$

use gradient descent:

$$w_n = w_{n-1} - \alpha_n \nabla V(f(x_n), y_n)$$

where α_n is the learning rate, and $V(f(x_i), y_i)$ a convex, differentiable loss function.

The learning can be done in batches, and the order of the input samples can be stochastic or deterministic.

Kernel methods: KRR (aka LS-SVM)

In kernel methods, the predictor at the n -th step has the form

$$f_n(\mathbf{x}) = \langle \mathbf{w}_n, \mathbf{x} \rangle = \sum_{k=1}^n (\alpha_n)_k \langle \mathbf{x}_k, \mathbf{x} \rangle = \sum_{k=1}^n (\alpha_n)_k K(\mathbf{x}_k, \mathbf{x})$$

where $\mathbf{w}_n = X_n^\top \alpha_n$, and $\alpha_n = (K(\mathbf{x}_n, \mathbf{x}_n) + \lambda I)^{-1} \mathbf{y}_n$.

Naive kernel regression: weights are recursively obtained using

$$\begin{aligned}\alpha_o &= 0 \\ (\alpha_n)_i &= (\alpha_{n-1})_i, \quad i = 1, \dots, n-1 \\ (\alpha_n)_n &= \gamma_n \left(y_n - \sum_{k=1}^{n-1} (\alpha_{n-1})_k K(\mathbf{x}_k, \mathbf{x}_n) \right)\end{aligned}$$

begin γ_n is the learning rate.